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Physiological and biochemical responses of *Thalassiosira weissflogii* (diatom) to seawater acidification and alkalization

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Increasing atmospheric pCO_2 leads to seawater acidification, which has attracted considerable attention due to its potential impact on the marine biological carbon pump and function of marine ecosystems. Alternatively, phytoplankton cells living in coastal waters might experience increased pH/decreased pCO_2 (seawater alkalization) caused by metabolic activities of other photoautotrophs, or after microalgal blooms. Here we grew *Thalassiosira weissflogii* (diatom) at seven pCO_2 levels, including habitat-related lowered levels (25, 50, 100, and 200 µatm) as well as present-day (400 µatm) and elevated (800 and 1600 µatm) levels. Effects of seawater acidification and alkalization on growth, photosynthesis, dark respiration, cell geometry, and biogenic silica content of *T. weissflogii* were investigated. Elevated pCO_2 and associated seawater acidification had no detectable effects. However, the lowered pCO_2 levels (25 $\sim$ 100 µatm), which might be experienced by coastal diatoms in post-bloom scenarios, significantly limited growth and photosynthesis of this species. In addition, seawater alkalization resulted in more silicified cells with higher dark respiration rates. Thus, a negative correlation of biogenic silica content and growth rate was evident over the pCO_2 range tested here. Taken together, seawater alkalization, rather than acidification, could have stronger effects on the ballasting efficiency and carbon export of *T. weissflogii*.

Keywords: acidification, alkalization, biogenic silica, growth, photosynthetic performance, *Thalassiosira weissflogii*

Introduction

Diatoms account for as much as 20% of global primary production and about 40% of marine primary production (Nelson *et al.*, 1995; Falkowski *et al.*, 2004). Their importance in the biological carbon cycle is especially remarkable in coastal waters. Nearly 75% of primary production in coastal zones is contributed by this group of phytoplankton (Nelson *et al.*, 1995). In addition, diatoms play a critical role in biogeochemical cycle of silicon due to the presence of silicified cell wall (Tréguer *et al.*, 1995). While diatoms are globally ubiquitous, they thrive in turbulent conditions, such as nutrient-rich coastal waters (Armbrust, 2009).

In coastal regions, seawater carbonate chemistry is impacted by multiple drivers, such as atmospheric CO_2 dissolution, upwelling, watershed processes, tidal cycles, and biota metabolism. Among these drivers, metabolic effects play a major role in impacting pH/pCO_2 in coastal ecosystems (Duarte *et al.*, 2013 and references therein). Climate change and anthropogenic stressors are expected to increase the frequency and abundance of eutrophication events, and this will trigger more phytoplankton blooms in coastal waters (Pesce *et al.*, 2018). Phytoplankton blooms could decrease pCO_2 to <50 µatm, and consequently increase pH to 9 or above, due to the uptake of CO_2 by photosynthesis (Liu *et al.*, 2017). Meanwhile, atmospheric pCO_2 has increased by 40% relative to pre-industrial levels as result of anthropogenic emissions (Howes *et al.*, 2015). This leads to decreased pH and CO_3^{2-} concentration, but increased aqueous CO_2 and HCO_3^- concentrations (i.e. ocean acidification).

Hence, coastal diatoms might experience both seawater acidification and alkalization within a single day.

Effects of elevated CO_2 and associated seawater acidification on diatoms have attracted considerable attention due to their critical role in the biological carbon pump and the biogeochemical cycle of silicon. Elevated CO_2 could downregulate the carbon concentrating mechanisms (CCMs) of diatoms (Gao and Campbell, 2014 and references therein). The saved energy expenditure due to downregulation of CCMs could stimulate growth and photosynthesis (Mackey *et al.*, 2015). A meta-analysis has shown that seawater acidification increased growth and photosynthesis of diatoms by 17 and 12%, respectively (Kroeker *et al.*, 2013). However, responses to seawater acidification might be species specific (Li *et al.*, 2016) and impacted by other drivers (King *et al.*, 2011; Gao *et al.*, 2012; Hoppe *et al.*, 2015; Sugie and Yoshimura, 2016). On the other hand, seawater acidification imposes additional energy costs for maintenance of intracellular pH homeostasis, as decreased seawater pH leads to decreased cytosolic pH (Rokitta *et al.*, 2012). Thus the responses of diatoms to seawater acidification depend on the balance of positive effects of elevated CO_2 and negative effects of lowered pH.

CO_2 accounts for less than 1% of dissolved inorganic carbon (DIC) at current seawater pH. This percentage continues to decline as pH increases. Moreover, CO_2 is the only substrate for the carboxylation reaction of Ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco). Half-saturation constants for CO_2 of diatom Rubisco range from 23 to 68 µM (Badger *et al.*, 1998; Young *et al.*, 2016), which are much higher than current CO_2 concentrations in seawater (~13 µM at 20°C). Thus, in the absence of CO_2 concentrating mechanisms (CCMs), growth of diatoms could be CO_2 limited at high pH/low pCO_2 (Riebesell *et al.*, 1993; Giordano *et al.*, 2005). *Thalassiosira weissflogii*, as a model diatom, has been suggested to possess a CCM based on C_4 (Reinfelder *et al.*, 2000) or C_3–C_4 intermediate photosynthesis (Roberts *et al.*, 2007). There is limited knowledge about the growth and photosynthesis responses of *T. weissflogii* to seawater alkalization, but varied results have been reported, with reports that growth was unaltered (Burkhardt *et al.*, 1999a; Milligan *et al.*, 2004; Reinfelder, 2012; Goldman *et al.*, 2017), or enhanced (Milligan *et al.*, 2009), by lowered pCO_2 compared to current or higher pCO_2 levels. Photosynthesis of *T. weissflogii* was unaffected over a wide range of pCO_2 levels (Burkhardt *et al.*, 2001; Goldman *et al.*, 2017), and was only depressed at the lowest pCO_2 after acclimation to pCO_2 levels for 2 days (Burkhardt *et al.*, 2001). However, two strains of *T. weissflogii* showed 60 and 26% lower growth rates at lowered pCO_2 than at present-day pCO_2 (King *et al.*, 2015). The negative effect of lowered CO_2 on this species has also been reported in other studies (Burkhardt *et al.*, 1999b; Hervé *et al.*, 2012; Hopkinson *et al.*, 2013; Wu *et al.*, 2014). This discrepancy in responses might be a consequence of the pCO_2 level tested, the pCO_2 manipulation method, or simply strain-specific responses.

For example, earlier studies on effects of seawater acidification relied on adjustment of seawater carbonate chemistry with HCl or NaOH (Burkhardt *et al.*, 1999a, b; Hervé *et al.*, 2012; Hopkinson *et al.*, 2013) while other work adjusted pCO_2 and pH by bubbling (Burkhardt *et al.*, 2001; Milligan *et al.*, 2004; 2009; Reinfelder, 2012; Wu *et al.*, 2014; King *et al.*, 2015; Goldman *et al.*, 2017). Although a previous study has demonstrated identical growth rates of *T. weissflogii* regardless of whether pCO_2/pH was manipulated by adding pH buffer, bubbling or using a pH-stat (Shi *et al.*, 2009), the different manipulation methods might yield inconsistent results in other physiological parameters. In addition, only one or two lowered pCO_2 levels were set in most prior studies. This prevents us from clarifying the pCO_2 threshold below which growth and photosynthesis of this model diatom are limited. In the present study, we grew *T. weissflogii* at seven pCO_2 levels, including habitat-related lowered (25, 50, 100, and 200 µatm), present-day, (400 µatm), and elevated levels (800 and 1600 µatm). Effects of seawater acidification and alkalization on growth, photosynthesis, dark respiration, cell geometry, and biogenic silica (BSi) were investigated.

Material and methods

Culture conditions

The diatom *Thalassiosira weissflogii* (CCMA 102), originally isolated from Daya Bay (China), was obtained from the Center for Collections of Marine Algae (CCMA) of the State Key Laboratory of Marine Environmental Science (Xiamen University). Cells were maintained at 20°C with a 12: 12 h light and dark cycle under 150 µmol photons m^{-2} s^{-1} of photosynthetically active radiation (PAR). Artificial seawater (Sunda *et al.*, 2005) used for cell culture was filter sterilized (0.2 µm, Whatman Polycap 75AS) and nutrients were added according to the recipe for f/2 medium (Guillard and Ryther, 1962). $NaHCO_3$ was not added when preparing artificial seawater. The target DIC value was then achieved by adding Na_2CO_3 and $NaHCO_3$ (for pCO_2 levels below 400 µatm) or aerating with elevated pCO_2 air (for pCO_2 levels above 400 µatm, see below for details). This seawater was then used for the pCO_2 treatments. Polycarbonate (PC) bottles (500 ml) were autoclaved before being used for cultures. Cultures were maintained in exponential growth phase in sealed PC bottles and were diluted every 48 h. The maximum cell concentration was controlled to be below 2500 cells ml^{-1} to minimize the impact of cell metabolism on seawater carbonate chemistry.

Experiment setup

T. weissflogii was acclimated to seven pCO_2 levels (25, 50, 100, 200, 400, 800, and 1600 µatm) for at least 15 generations before final sampling. For the lowered pCO_2 levels (25, 50, 100, and 200 µatm), Na_2CO_3 and $NaHCO_3$ were added to DIC-free artificial seawater to obtain target pCO_2 values. The amounts of Na_2CO_3 and $NaHCO_3$ were calculated by the following equations: $X + Y$ = Target DIC; $X + 2 * Y + K$ = TA, where X and Y indicate the concentrations (µmol kg^{-1}) of $NaHCO_3$ and Na_2CO_3 in artificial seawater respectively; K is the alkalinity contributed by ions besides HCO_3^- and CO_3^{2-}; TA represents the total alkalinity (TA) in artificial seawater. Here, target DIC was calculated using the CO2SYS programme (version 2.1 by E Lewis and D W R Wallace) based on target pCO_2 level and TA; TA was set as 2300 µmol kg^{-1}, which is the approximate mean value of artificial seawater. In this way, DIC of the seawater was adjusted to the target value and TA remained constant. The elevated pCO_2 levels (800 and 1600 µatm) were achieved using a CO_2 plant growth chamber (HP1000G-D, RuiHua), in which air and pure CO_2 were mixed to the desired pCO_2. The media were pre-aerated with CO_2-enriched air before cell inoculation (cultures were not aerated during incubation). For the ambient pCO_2 level (400 µatm), both methods mentioned above were applied. Three independent cultures were used for each treatment and cells exposed to the different treatments were cultured in one growth chamber (GXZ, JiangNan). PC bottles were randomly placed in positions with the same level of PAR in the chamber to avoid artefacts.

Seawater carbonate chemistry

Samples for TA determination were filtered through 0.22 μm polyethersulfone membranes (Millipore Express) after 48 h culturing. TA was determined on 25 ml samples using the Gran titration method (Dickson *et al.*, 2007). The pH_{NBS} values were measured by a three-point calibrated pH meter (FE20, Mettler Toledo). DIC and CO_2 in seawater samples were calculated using the CO2SYS program based on pH and TA.

Cell concentration, volume, and surface area

Cell concentration was determined with an optical microscope (DM500, Leica) using 0.1 ml plankton chamber (DSJ-01, Dengxun). Specific growth rate was calculated as the ratio of ln (N_2/N_1) and the time between dilution (d), where N_2 and N_1 represent the cell concentrations before and after the dilution, respectively. Cell width (d) and length (h) were determined with the microscope and ToupView software. At least 15 cells from each replicate were measured for cell size. Cell volume (V) and surface area (SA) were estimated using the equations for a cylinder: $V = \pi * (d/2)^2 * h$; $SA = \pi * d * (h + d/2)$.

Chlorophyll *a* and biogenic silica

Cells were harvested on GF/F filters (25 mm, Whatman) and then extracted with 100% methanol overnight at 4°C for analysis of chlorophyll *a* content. After centrifugation (5000 g for 10 min), the absorption of the supernatant was determined spectrophotometrically (Lamda35, Perkin Elmer) at 632, 665, and 750 nm. Chlorophyll *a* content was calculated according to Ritchie (2006). For measurement of BSi, cells were harvested on 0.45 μm mixed cellulose ester filters (25 mm, Xinya). Filters with cells were dried at 80°C for 24 h. Then cells were digested by 0.2 mol l^{-1} NaOH for 40 min and neutralized with 1 mol l^{-1} HCl following the method of Brzezinski and Nelson (1995). The absorption of the samples was determined at 810 nm by a spectrophotometer (Lamda35, Perkin Elmer) after reacting with ammonium molybdate and the reducing agent.

Chlorophyll fluorescence

For cells grown at ambient and elevated pCO_2 levels (400, 800, and 1600 μatm, i.e. pCO_2 adjusted by aeration), chlorophyll fluorescence parameters were determined using an AquaPen-C fluorometer (AP-C100, Photon Systems Instruments). However, due to instrument malfunction, a Water-PAM (Walz) was used for the equivalent measurements of cells in lowered and ambient pCO_2 treatments (25, 50, 100, 200, and 400 μatm). For maximum quantum yield of PSII (F_v/F_m) analysis, samples were dark adapted for 10 min before measurement. Then minimum (F_0) and maximum chlorophyll fluorescence (F_m) for dark-adapted samples were determined and F_v/F_m was calculated as: $F_v/F_m = (F_m - F_0)/F_m$. Rapid light curves (RLCs) were conducted to establish the relationship between relative electron transport rate (rETR) and light intensity. A range of progressively increasing actinic light intensities (10, 20, 50, 100, 300, 500, and 1000 μmol photons m^{-2} s^{-1} for the AquaPen-C fluorometer and 25, 81, 120, 184, 272, 415, 619, 878, and 1218 μmol photons m^{-2} s^{-1} for the Water-PAM) were used in the RLC measurements. RLCs were fitted to the following model: $rETR = PAR/(a * PAR^2 + b * PAR + c)$, where PAR is the photon flux density of actinic light (μmol photons m^{-2} s^{-1}), a, b, and c are model parameters. Relative maximum electron transport rate ($rETR_{max}$), apparent photon transfer efficiency (α), and light saturation point (I_k) were calculated from a, b, and c according to the equations in Eilers and Peeters (1988).

Photosynthesis and dark respiration

A Clark-type oxygen electrode (5300A, YSI) was used to measure photosynthetic oxygen evolution and dark respiration rate. Samples of 100 ml from each culture replicate were gently filtered onto 0.45 μm mixed cellulose ester filters (50 mm, Xinya). Subsequently, cells were re-suspended into 10 ml filtered seawater (carbonate chemistry was adjusted by aeration or adding Na_2CO_3 and $NaHCO_3$, as mentioned above). Subsamples of 8 ml were injected into the oxygen electrode chamber and were magnetically stirred. The temperature in the oxygen electrode chamber was controlled at 20°C by a refrigerated circulating bath (DHX-2005, Xianou). A halogen lamp was used for the photosynthetic oxygen evolution determination, and the light intensity was set as 150 μmol photons m^{-2} s^{-1} (the same as culture light intensity). Oxygen evolution and consumption rates were recorded when they became constant (∼6 min). The remaining 2 ml sample was used to determine the cell concentration microscopically.

Statistical analysis

All data in the present study are reported as the mean ± standard deviation (SD) of triplicate independent samples ($n = 3$), except photosynthesis, and respiration rates in cells grown at 25 μatm pCO_2 (where only two replicates were determined due to improper manipulation). For chlorophyll fluorescence parameters, the percentage changes relative to the mean value for the 400 μatm treatment, rather than raw data, are presented since different fluorometers were used (as mentioned above). Shapiro–Wilk and Levene tests were used to test the normality and equal variance of data, respectively. Significant differences among treatments were tested using one-way analysis of variance (ANOVA) and *post hoc* Duncan test with a significance level of $p < 0.05$. When assumptions for parametric statistical test were not met, the Kruskal–Wallis nonparametric test was used to analyze the significant difference.

Results

The carbonate chemistry parameters are shown in Table 1. pH decreased from 9.02 ± 0.01 to 7.63 ± 0.01 as pCO_2 increased from 25 to 1600 μatm, and TA was unaltered. DIC, HCO_3^-, and aqueous CO_2 concentrations increased with increased pCO_2. For the 400 μatm pCO_2 level, there were no differences in these parameters between the two CO_2 manipulation methods (aeration and chemical additions).

The specific growth rate of *T. weissflogii* significantly decreased with increased pH, but the rate was enhanced with increased CO_2 and HCO_3^- concentrations (Figure 1, $p < 0.001$, one-way ANOVA). Cells grown at 50 μatm pCO_2 showed 20% higher specific growth rate relative to cells grown at 25 μatm pCO_2, and a 17% enhancement in the rate was observed when pCO_2 level increased from 200 to 400 μatm. There was no change in growth rate in cells grown at 800 and 1600 μatm pCO_2 levels relative to 400 μatm. No impact of CO_2 manipulation methods on growth rate was observed in cells grown at 400 μatm pCO_2. Elevated pCO_2 from 25 to 400 μatm enhanced cell volume and surface

Table 1. Carbonate chemistry parameters of culture media.

Treatments (μatm)	pH_{NBS}	TA (μmol kg^{-1})	DIC (μmol kg^{-1})	HCO_3^- (μmol kg^{-1})	CO_2 (μmol kg^{-1})
25	9.02 ± 0.01^a	2364 ± 19^a	1391 ± 11^a	787 ± 11^a	0.8 ± 0^a
50	8.83 ± 0.01^b	2376 ± 24^a	1555 ± 13^b	1037 ± 4^b	1.6 ± 0^b
100	8.63 ± 0.01^c	2402 ± 39^a	1752 ± 30^c	1335 ± 22^c	3.4 ± 0.1^c
200	8.44 ± 0.02^d	2374 ± 38^a	1880 ± 31^d	1561 ± 28^d	6.1 ± 0.3^d
400	8.17 ± 0.02^e	2381 ± 23^a	2062 ± 30^e	1847 ± 32^e	13.2 ± 0.7^e
400*	8.16 ± 0.01^e	2365 ± 7^a	2052 ± 8^e	1842 ± 8^e	13.5 ± 0.3^e
800	7.93 ± 0.01^f	2366 ± 33^a	2172 ± 33^f	2021 ± 32^f	25.5 ± 0.7^f
1600	7.63 ± 0.01^g	2365 ± 28^a	2287 ± 30^g	2165 ± 28^g	54.6 ± 1.4^g

"400" and "400*" represent ambient pCO_2 level adjusted by adding chemicals and bubbling, respectively. Values of carbonate chemistry parameters are means ± SD of triplicate cultures. The different letters indicate significant ($p < 0.05$) differences among treatments.

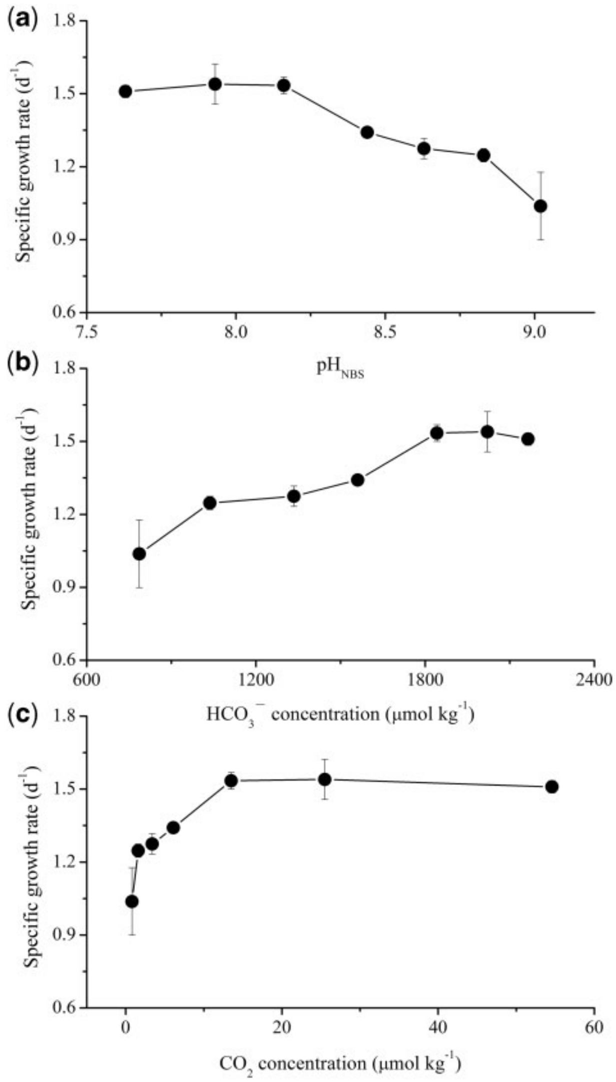

Figure 1. Specific growth rates (d^{-1}) of *T. weissflogii* cells at different pH (a), HCO_3^- (b), or CO_2 concentration (c). The data are means ± SD of triplicate cultures ($n = 3$). For cells grown at 400 μatm pCO_2, there were no differences in specific growth rate between the two CO_2 manipulation methods (aeration and chemical additions). For simplicity, only ambient pCO_2 level adjusted by bubbling was shown here.

area by 77 and 45% respectively (Table 2, $p < 0.001$ for cell volume, one-way ANOVA; $p = 0.004$ for surface area, Kruskal–Wallis test), and cells grown at 800 and 1600 μatm pCO_2 were 14% smaller than cells at 400 μatm. Decreased SA: V ratios were observed as pCO_2 increased from 25 to 400 μatm, then the ratio increased slightly at 800 and 1600 μatm pCO_2 ($p < 0.001$, one-way ANOVA).

The chlorophyll *a* content was relatively constant over the pH, CO_2, and HCO_3^- ranges tested here. A significant difference was only observed in cells grown at 25 and 50 μatm pCO_2, with 33% lower chlorophyll *a* content compared to other treatments (Figure 2a–c, $p = 0.024$, Kruskal–Wallis test). An increased trend was observed in BSi content as pH increased and opposite trends were found as CO_2 and HCO_3^- concentrations increased (Figure 2d–f, $p = 0.003$, Kruskal–Wallis test), but there was no change in BSi content when pCO_2 increased from 400 to 1600 μatm. Similarly, elevated pCO_2 decreased BSi per surface area as shown in BSi content per cell (Table 2, $p < 0.001$, one-way ANOVA).

No significant differences in F_v/F_m and α were found among treatments (Figure 3a and c, $p = 0.135$ and 0.430, respectively, Kruskal–Wallis test). For cells grown at 25 μatm pCO_2, rETRmax was 44% lower compared to that from cells at 400 μatm, but the value was similar in all other treatments (Figure 3b, $p = 0.001$, one-way ANOVA). Cells grown at 25 and 50 μatm pCO_2 showed 43 and 22% lower I_k values relative to other treatments, respectively (Figure 3d, $p < 0.001$, one-way ANOVA).

While elevated pCO_2 from 25 to 100 or from 200 to 1600 μatm showed no impact on photosynthetic oxygen evolution rate, an overall increase in the rate was observed with increased CO_2 and HCO_3^- concentrations (and decreased pH). The rate increased by 50% as pCO_2 increased from 100 to 200 μatm (Figure 4a–c, $p = 0.012$, Kruskal–Wallis test). Over the pCO_2 range from 25 to 100 μatm, dark respiration rate decreased with increased CO_2 and HCO_3^- concentrations, but increased with increased pH, though no further change was observed from 100 to 800 μatm (Figure 4d–f, $p = 0.013$, Kruskal–Wallis test). For cells grown at 400 μatm pCO_2, there were no differences in oxygen evolution and consumption rates between the two CO_2 manipulation methods (aeration and adding chemicals).

Discussion

Our results clearly showed that seawater alkalization, rather than acidification, strongly impacted the physiological and

Table 2. Cell volume (μm^3), surface area (μm^2), surface area-to-cell volume ratio, and BSi content per surface area (fmol μm^{-2}) of *T. weissflogii* cells grown at different pCO_2 levels.

Treatments (μatm)	Cell volume (μm^3)	Surface area (μm^2)	SA: V	BSi (fmol μm^{-2})
25	1211 ± 89^a	638 ± 31^a	0.53 ± 0.01^a	7.8 ± 2.6^a
50	1553 ± 124^b	749 ± 42^b	0.48 ± 0.01^b	6.5 ± 0.6^{ab}
100	1734 ± 70^c	806 ± 22^c	0.46 ± 0.01^c	5.2 ± 0.6^b
200	1817 ± 50^c	831 ± 14^c	0.46 ± 0.01^c	2.7 ± 0.1^c
400	2141 ± 36^d	922 ± 10^d	0.43 ± 0.01^d	1.1 ± 0.2^c
400*	2074 ± 169^d	907 ± 50^d	0.44 ± 0.01^d	1.2 ± 0.2^c
800	1835 ± 73^c	834 ± 22^c	0.45 ± 0.01^c	0.9 ± 0.2^c
1600	1845 ± 81^c	837 ± 24^c	0.45 ± 0.01^c	0.9 ± 0.1^c

"400" and "400*" represent ambient pCO_2 level adjusted by adding chemicals and bubbling, respectively. Values are means ± SD of triplicate cultures. The different letters indicate significant ($p < 0.05$) differences among treatments.

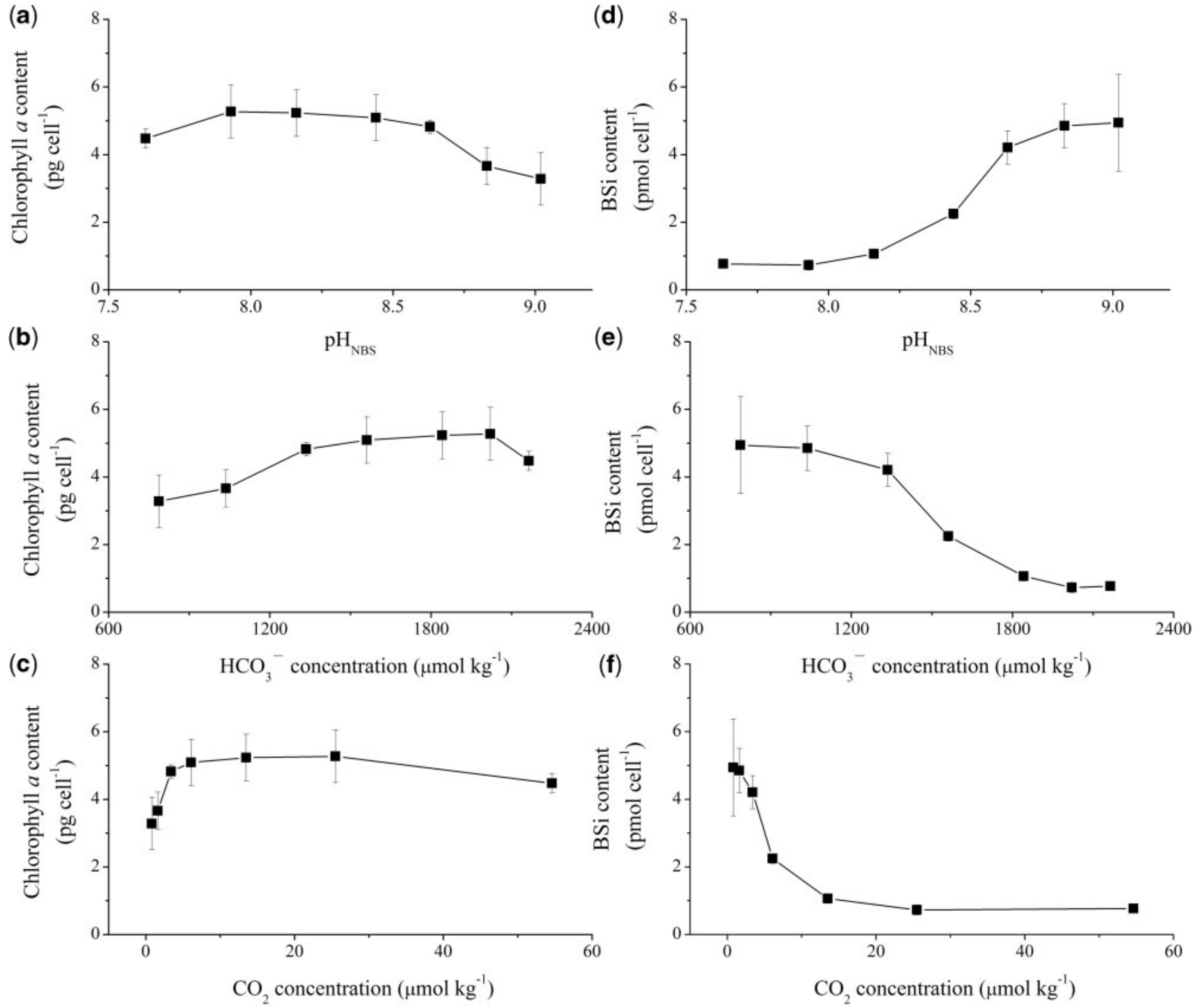

Figure 2. Chlorophyll *a* (pg $cell^{-1}$) (a, b, and c) and BSi content (pmol $cell^{-1}$) (d, e, and f) of *T. weissflogii* cells at different pH (a and d), HCO_3^- (b and e), or CO_2 concentration (c and f). The data are means ± SD of triplicate cultures ($n = 3$). For cells grown at 400 μatm pCO_2, there were no differences in chlorophyll *a* and BSi contents between the two CO_2 manipulation methods (aeration and chemical additions). For simplicity, only ambient pCO_2 level adjusted by bubbling was shown here.

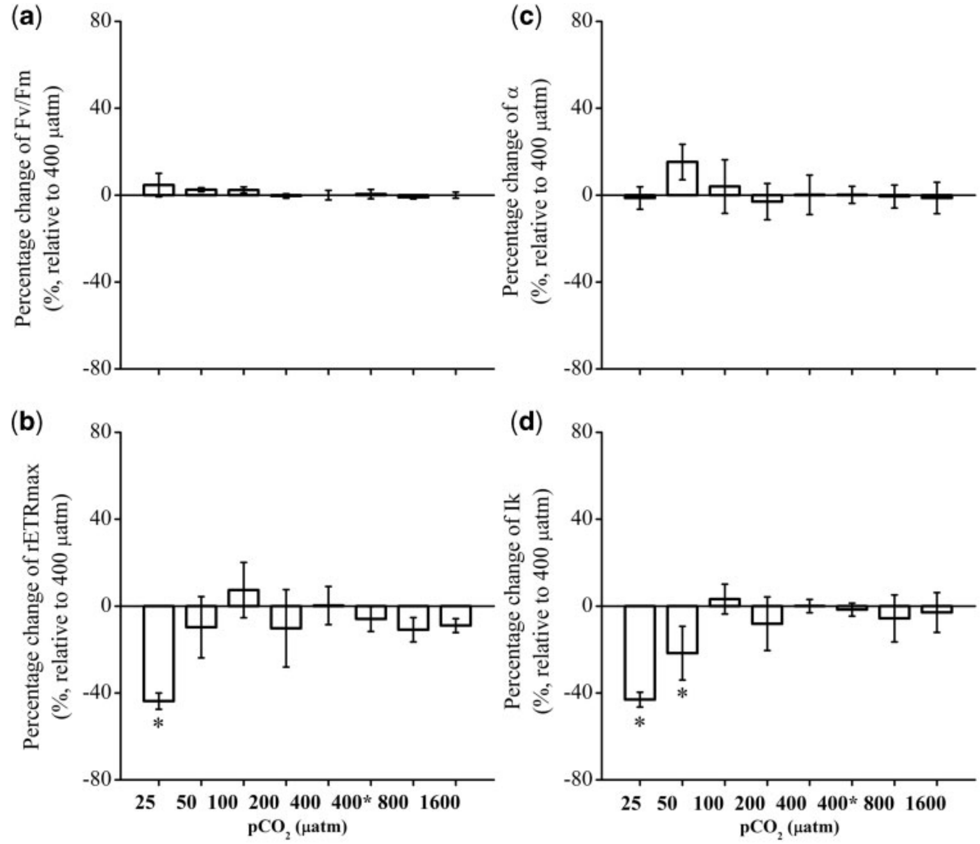

Figure 3. Percentage changes of maximum photochemical quantum yields (F_v/F_m, a), relative maximum electron transport rate ($rETR_{max}$, b), apparent photon transfer efficiency (α, c), and light saturation point (I_k, d) (%) of *T. weissflogii* cells grown at different pCO_2 levels (relative to 400 μatm treatment). "400" and "400*" represent ambient pCO_2 level adjusted by adding chemicals and bubbling, respectively. The mean values of these parameters of cells grown at 400 μatm pCO_2 (zero line) were as follows: F_v/F_m: 0.57 ± 0.01 (Water-PAM), 0.70 ± 0.01 (AquaPen-C fluorometer); $rETR_{max}$: 116 ± 10 (Water-PAM), 87 ± 5 (AquaPen-C fluorometer); α: 0.22 ± 0.02 (Water-PAM), 0.29 ± 0.01 (AquaPen-C fluorometer); I_k: 511 ± 15 (Water-PAM), 300 ± 9 μmol photons m^{-2} s^{-1} (AquaPen-C fluorometer).The data are means ± SD of triplicate cultures ($n = 3$). Asterisks indicate significant ($p < 0.05$) differences between pCO_2 treatments.

biochemical performance of *T. weissflogii*. Cells showed slower growth and photosynthetic rates with higher respiration at low pCO_2 levels, conditions which might be experienced by coastal phytoplankton in post-bloom scenarios. In addition, cells were more silicified at low pCO_2 levels, which might increase the sinking velocity of organic matter. These results suggest that coastal diatoms might be insensitive to seawater acidification, whereas increased pH and reduced CO_2 availability (seawater alkalization) could markedly alter the contribution of diatoms to the local biological carbon pump.

The different CO_2 manipulation methods (aeration and adding chemicals) showed no detectable impact on physiological parameters tested here. Bubbling seawater with premixed gas exactly simulates the changes in carbonate chemistry predicted to occur by the end of this century (changing DIC at constant TA) (Gattuso *et al.*, 2010). Adding Na_2CO_3 and $NaHCO_3$ to DIC-free artificial seawater in the present study yields the same carbonate chemistry conditions as aeration (Table 1). Thus the target DIC could be easily achieved by adding specific amounts of Na_2CO_3 and $NaHCO_3$ without alteration of TA and could be an alternative way to manipulate pH/pCO_2 in studies on responses of phytoplankton cells to seawater acidification, especially for species that are negatively affected by bubbling. However, it should be kept in mind that this method is applicable only when phytoplankton biomass is low, and the containers used for culturing must be sealed to avoid gas exchange.

Growth and photosynthesis of *T. weissflogii* were depressed by limited CO_2 availability and/or increased pH over the pCO_2 range from 25 to 100 μatm (CO_2: 0.8 ~ 3.4 μmol kg^{-1}; pH_{NBS}: 8.63 ~ 9.02), as reflected by enhanced rates with increased pCO_2. The 23% higher SA: V ratio in cells grown at low pCO_2 might partly alleviate the CO_2 limitation, as the higher ratio presumably facilitates CO_2 diffusion. At low pCO_2 levels, bicarbonate might be the major DIC species taken up by *T. weissflogii* cells. This species could utilize CO_2 and HCO_3^- at approximately the same rate at current pCO_2 level, and a decrease in the CO_2: HCO_3^- uptake ratio was observed at low pCO_2 (Burkhardt *et al.*, 2001). Marine diatoms are suggested to utilize bicarbonate indirectly through the activity of extracellular carbonic anhydrase (CA) (Martin and Tortell, 2008) or directly via specific plasmalemma-located transporters (Nakajima *et al.*, 2013), though *T. weisflogii* appears use a biochemical rather than a biophysical CCM (Reinfelder *et al.*, 2000; Roberts *et al.*, 2007). Both extracellular and intracellular CA activities of *T. weissflogii* were markedly enhanced by low pCO_2 and half-saturation concentration of DIC for photosynthesis was decreased with decreasing pCO_2 (Burkhardt *et al.*, 2001),

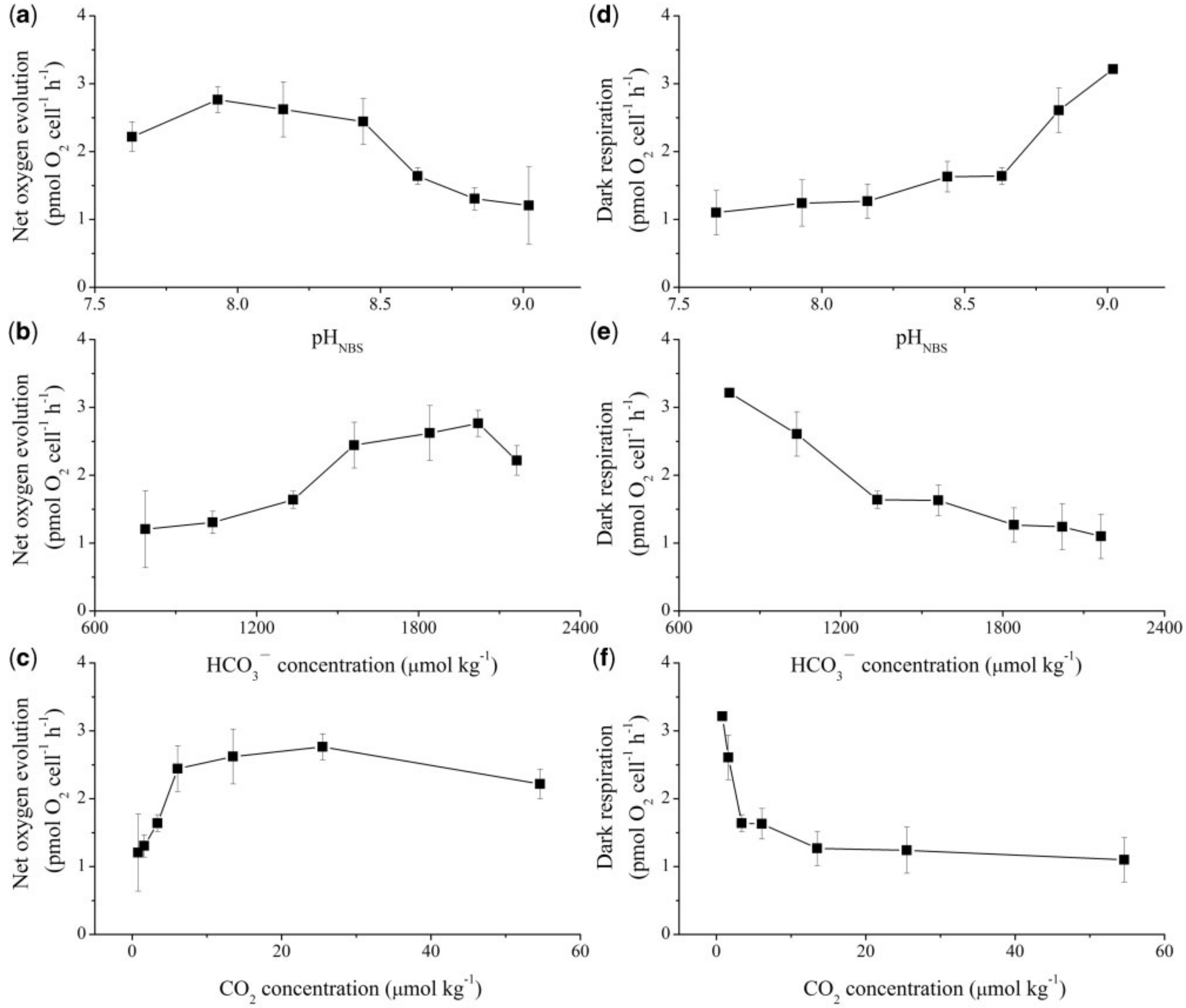

Figure 4. Net oxygen evolution (a, b, and c) and dark respiration rates (pmol O_2 cell^{-1} h^{-1}) (d, e, and f) of *T. weissflogii* cells at different pH (a and d), HCO_3^- (b and e), or CO_2 concentration (c and f). The data are means ± SD of triplicate cultures ($n = 3$), except for cells grown at 25 μatm pCO_2 ($n = 2$). For cells grown at 400 μatm pCO_2, there were no differences in net oxygen evolution and dark respiration rates between the two CO_2 manipulation methods (aeration and chemical additions). For simplicity, only ambient pCO_2 level adjusted by bubbling was shown here.

indicating that CCMs of *T. weissflogii* were upregulated under such conditions.

Mitochondrial respiration provides energy and carbon skeletons for cell biosynthesis and maintenance. Hence, enhanced respiration could theoretically lead to increased growth rate, as shown in diatom and other microalgae cultured under optimal light and nitrate conditions (Geider and Osborne, 1989; Li *et al.*, 2018). However, various relationships may be observed, since growth is a proxy for overall cell metabolism. Enhanced respiration may lead to increased (Wu *et al.*, 2010), unaltered (Yang and Gao, 2012), or depressed diatom growth. For instance, a negative relationship between dark respiration and growth was observed in the present study (Figure 5). Interestingly, a negative relationship was also detected in the diatom *Thalassiosira pseudonana* cultured under nitrate-limited conditions (Li *et al.*, 2018). Enhanced respiration seems to be a strategy for cells to cope with stressful (CO_2- or nitrate-limited) conditions. Both mitochondrial respiration and photorespiration decrease the efficiency of photosynthetic carbon fixation in photoautotrophs (Raven and Beardall, 2005). Rubisco catalyzes both carboxylation and oxygenation reactions. The relative activities of these two competitive reactions depend on the Michaelis constant for CO_2 and O_2, maximum carboxylation and oxygenation rates, and the concentrations of aqueous CO_2 and dissolved O_2 (Farquhar *et al.*, 1980). Thus, photorespiratory activity of Rubisco might be theoretically enhanced at lowered pCO_2 levels, as shown by the decreased Rubisco carboxylation: oxygenation ratio when aqueous CO_2 decreased (Reinfelder, 2011). However, marine phytoplankton has evolved CCMs to overcome the low affinity for CO_2 of Rubisco and relative low aqueous CO_2 in seawater (Giordano *et al.*, 2005). Hence, effects of lowered CO_2 on photorespiration and net photosynthesis are complicated: on one hand, lowered aqueous CO_2

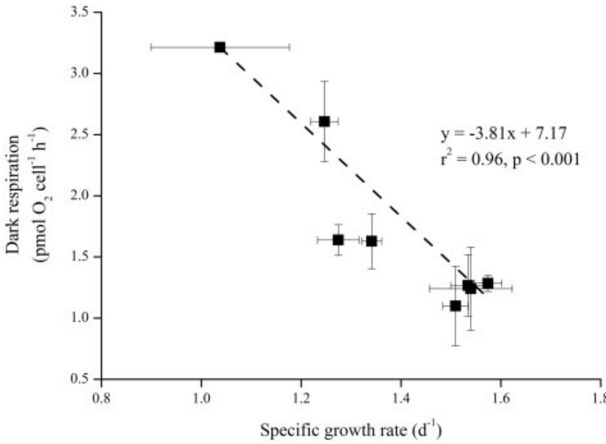

Figure 5. Relationships between dark respiration rates (pmol O_2 cell^{-1} h^{-1}) and specific growth rates (d^{-1}) of *T. weissflogii* cells grown at different pCO_2 levels. Linear regression is indicated by dashed line.

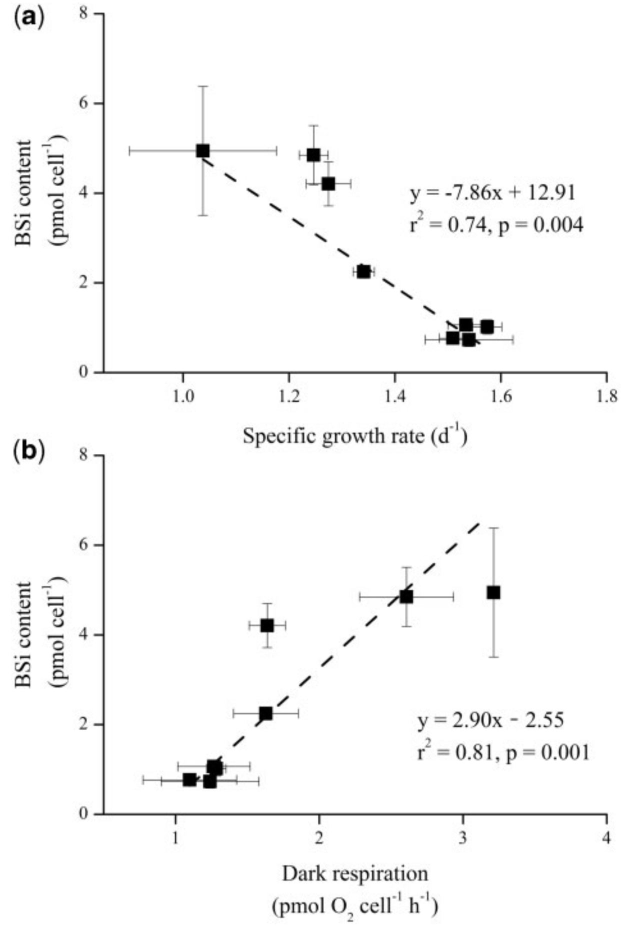

Figure 6. Relationships between BSi content (pmol cell^{-1}) and specific growth rates (d^{-1}) (a) or dark respiration rates (pmol O_2 cell^{-1} h^{-1}) (b) of *T. weissflogii* cells grown at different pCO_2 levels. Linear regressions are indicated by dashed lines.

increases the relative ratio of dissolved O_2 to CO_2; on the other hand, lowered CO_2 would upregulate CCMs, which would increase CO_2 concentration in the proximity of Rubisco and might suppress photorespiration. More energy would be invested in DIC acquisition when CCMs are upregulated, thus other aspects of cell metabolism might be depressed. This might be partly responsible for the depressed growth with decreasing CO_2 in the present study.

The silicified cell walls in diatoms provide effective mechanical protection and can act as armour against predators (Hamm *et al.*, 2003). Furthermore, the silica frustule has been suggested act as an effective pH buffer for the catalytic activity of CA, which catalyzes the conversion of bicarbonate to CO_2 (Milligan and Morel, 2002). Thus, cells might be more silicified at low pCO_2 levels, when phytoplankton cells are limited by CO_2 availability. Indeed, a negative correlation of BSi content and growth rate was evident over the pCO_2 range tested here (Figure 6a). Milligan *et al.* (2004) also found an enhanced BSi quota in *T. weissflogii* cells grown at lowered pCO_2 level, which resulted from a lower Si efflux: influx ratio (i.e. greater retention of Si). Besides CO_2 limitation, light (Liu *et al.*, 2016), iron (Hutchins and Bruland, 1998), and nutrient limitation (Claquin *et al.*, 2002) were also shown to increase BSi content with decreasing growth rate. This trend could be attributed to the G2 + M elongation of cells with decreased growth rate (Claquin *et al.*, 2002). Mitosis and cell wall synthesis, and thus the major uptake of silicon, occur during the G2 + M phase (Martin-Jézéquel *et al.*, 2000). In the present study, the BSi content of *T. weissflogii* cells correlates positively with dark respiration (Figure 6b). The increased silicification might partly be responsible for enhanced dark respiration at low pCO_2, since aerobic respiration, rather than photosynthesis, has been suggested as the source of energy required for silicification (Lewin, 1955; Sullivan, 1976). Based on these results, the more silicified cell wall at low pCO_2 levels might increase the sinking velocity of organic matter, which can then be degraded by microbes. This would lead to lower oxygen and increased acidity in subsurface waters (Melzner *et al.*, 2013). Hence, in post-bloom conditions, highly silicified diatom cells might accelerate the occurrence and development of hypoxia and acidification, especially in coastal waters.

The elevated pCO_2 levels (800 and 1600 μatm) had no detectable effects on growth, chlorophyll and BSi content, photosynthesis, and respiration of *T. weissflogii* relative to the current pCO_2 (400 μatm). Growth of this species was thus insensitive to seawater acidification, as reported in previous studies (Reinfelder, 2012; King *et al.*, 2015; Passow and Laws, 2015; Li *et al.*, 2016; Goldman *et al.*, 2017; Yang *et al.*, 2018). In contrast, pCO_2 close to pre-industrial levels (200 μatm) slightly depressed the growth and cell volume, but increased the BSi quota. Even lower pCO_2 levels (25 ∼ 100 μatm), which might be experienced by coastal phytoplankton in post-bloom scenarios, significantly limited growth and photosynthesis of this species. Depressed growth at lowered pCO_2 (aqueous $CO_2 < 10\,\mu mol\ kg^{-1}$) was also observed for a range of diatoms (Burkhardt *et al.*, 1999b). Based on these results, *T. weissflogii* cells will not be impacted by elevated pCO_2 and associated seawater acidification, but CO_2 limitation and/or increased pH caused by seawater alkalization might depress its growth and carbon fixation. In addition, seawater alkalization leads to more silicified (thicker frustule), and possibly faster-sinking cells. These changes in the ballasting efficiency and carbon export would have important implications for local carbon and silicon budgets.

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